Theoretical investigation of Dosimeter accuracy for linear energy transfer measurements in proton therapy: A comparative study of the stopping power ratios

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Abstract

Background: Accurate stopping power data is crucial for proton therapy dosimetry to ensure precise radiation dose delivery.

Purpose: To evaluate the accuracy of dosimetry materials in measuring linear energy transfer (LET).

Methods: High-Z_{eff} (BaFBr) and low-Z_{eff} (Al2O3, water) materials were characterized using Bethe-Bloch theory and semiempirical models. The stopping power and ratios were calculated for proton energies ranging from 0.01 to 10,000 MeV. Results of stopping powers were compared with calculations from SRIM-2008 and PSTAR data for validation.

Results: The water-to-air ratio was approximately 1.05 at 10 MeV, the water-to-Al2O3 ratio was approximately 1.12 at 1 MeV, and the BaFBr-to-water ratio approximately 0.95. This result confirms the accurate LET measurements over various proton energies.

Conclusion: This study validates the use of both high-Z_{eff} and low-Z_{eff} materials for accurate LET measurements in proton therapy, thereby enhancing dosimetry precision and potentially improving radiation therapy outcomes.

Keywords:

Linear Energy Transfer, Dosimetry, Bethe-Bloch Theory, Proton Therapy, Stopping Power (Ratios)

1. Introduction

Accurate data in proton therapy dosimetry are crucial for optimizing treatment efficacy and minimizing damage to surrounding healthy tissues. Proton therapy is favored for its precise dose distribution, but this precision hinges on accurate measurements of linear energy transfer (LET), which describes the energy deposited by protons as they traverse biological tissues. Discrepancies in LET measurements can lead to significant variations in treatment outcomes, highlighting the importance of validating dosimetric materials used in clinical settings.

Previous studies have explored various storage phosphor materials for dosimetry ranging from high atomic number (high-Z_{eff}) materials such as $BaFBrI{:}\,Eu^{2+}$ to low-Z_{eff} materials such as $KCI{:}\,Eu^{2+}$, their study suggests that the high-Z_{eff} storage phosphor could be used in conjunction with a low-Z_{eff} $KCI{:}\,Eu^{2+}$ for linear energy transfer (LET) and simultaneous proton dose measurements [1]. However, a study by [2] has shown that the essential optical and dosimetric properties of europium-doped potassium chloride (KCL) as a storage phosphor can be characterized for precise proton dosimetry, and its dose radiation response can be effectively modeled using an analytical radiation transport model. It is important to use Monte Carlo (MC) methods to model $KCI{:}\,Eu^{2+}$ point dosimeters and to investigate the dose responses of two-dimensional (2D) $KCI{:}\,Eu^{2+}$ storage phosphor films (SPFS) [3].

These studies have provided foundational knowledge on the stopping powers and ratios of these materials across different energy ranges. However, limitations remain, particularly regarding the consistency and accuracy of LET measurements at lower proton energies. For instance, while high-Z_{eff} materials offer high sensitivity, their accuracy at clinically relevant energy ranges has been questioned. Conversely, low-Z_{eff} materials like water are widely used because of their tissue equivalence, but they may lack the necessary resolution for precise LET measurements at lower energies.

In addition, other authors attempted to create 2D $KCI{:}\,Eu^{2+}$ storage phosphor films (spfs) via physical vapor deposition (PVD) and tape casting [4]. A recently proposed storage phosphor material for radiation dosimetry investigates the radiation hardness of europium-doped potassium chloride ($KCI{:}\,Eu^{2+}$) [5]. However, [6] proved that $KCI{:}\,Eu^{2+}$ dosimeters possess many preferable dosimetric features, which are favorable for radiation therapy dosimetry. A recent review investigated how gel dosimetry can be used in low energy procedures like radiotherapy and interventional radiology [7]. The precision of LET measurements using a low-Z_{eff} material such as Al2O3:C OSLDs in proton beams by utilizing current, automated readers capable of conducting POSL measurements and irradiation [8]. In addition, [9] demonstrated and assessed the feasibility of calibrating Al2O3:C OSLDs for LET measurements using novel approaches.

Recent research has updated effective energies for reducing range error in proton treatment and ion beam therapy using helium, carbon, oxygen, and neon ions, as well as elemental values I [10]. They evaluated the impact of these updates on the stopping power ratio (SPR) estimation in computed tomography-based treatment planning [10]. The modifications increased stopping power ratios (SPRs) in soft tissues by up to 0.5% and decreased stopping power ratios (SPRs) in bone tissues by up to 1.9% compared with existing clinical settings [10]. An analysis of the efficacy of spectral computed tomography (CT) employing a fast kV-switching dual-energy technique to predict the stopping power ratio (SPR) for proton radiotherapy was conducted [11].

Therefore, [12] has used an experiment to confirm Geant4's proton physics models (version 10.6). Eleven different inserts, consisting of acrylic, delrin, high-density polyethylene, and polytetrafluoroethylene, were examined to determine their stopping power ratios (SPRs) using a superconducting synchrocyclotron, which produces a scattering proton beam [12]. On the contrary, research on dosimetry in the low-energy range employed in clinical settings for diagnostic or therapeutic purposes has disclosed various methods rooted in fundamental quantum mechanics that can be advantageous for handling low-energy interactions [13]. In the realm of medical physics, precise electronic stopping power data are of paramount importance for computing radiation-induced consequences in a multitude of proton therapy dosimetry applications, as previously documented [18-21].

In this study, the aim was to evaluate the accuracy of dosimetry materials in measuring LET for proton therapy, by covering a wide spectrum of proton energies and validating theoretical models with Calculations from PSTAR and simulation data (SRIM-2008), thereby enhancing the reliability of LET measurements in clinical proton therapy dosimetry.

2. Methods

2.1 Study Design and Participants

This study was designed to evaluate the accuracy of different dosimetric materials in measuring linear energy transfer (LET) across proton energies. The study includes both high-Z_{eff} materials, such as Barium Fluorobromide (BaFBr), and low-Z_{eff} materials, such as Aluminum Oxide (Al2O3) and water. These materials were selected due to their relevance in clinical proton therapy and their differing physical properties, which provide a broad basis for comparison.

Stopping power ratios were calculated using the PSTAR database for proton energies ranging from 0.01 MeV to 10,000 MeV. We applied the Bethe-Bloch theory with density and shell corrections for high-energy protons and a semiempirical model for low-energy protons for stopping power calculation. We validated our theoretical stopping power data by comparing them with the calculations from SRIM and PSTAR data for materials like aluminum, Al2O3, BaFBr, and oxygen.

Material Selection

High-Z_{eff} Material: No physical reagent of BaFBr was used. All calculations were theoretical and based on the known chemical properties of Barium Fluorobromide (BaFBr)

- **Relevance**: BaFBr is known for its high sensitivity and is commonly used in dosimetry because of its high atomic number, which enhances its interaction with protons.

Low-Z_{eff} Materials: Aluminum Oxide (Al2O3) and Water

- **Aluminum Oxide (Al2O3)**:
  - **Relevance**: Al2O3 was chosen for its intermediate atomic number, its wide usage in dosimetry, and its reliable physical properties.
- **Water**:
  - **Representation**: Water is used as a reference material because of its tissue equivalence, making it a standard for dosimetry studies.

2.2 Computational Methods

Theoretical calculations were performed using the Bethe-Bloch theory, incorporating density effect and shell corrections for high-energy protons, and a semiempirical model for low-energy protons. Calculation validation was conducted using data from the SRIM-2008 and PSTAR databases. **Bethe-Bloch Theory**: Applied for high-energy protons with necessary corrections. **Semi-Empirical Model**: Used for low-energy protons to enhance accuracy. **SRIM Data**: Provided stopping power and range data calculations for various materials. **PSTAR Data**: Offered validation of stopping power data calculations for protons in water and other reference materials. **Barium Fluorobromide (BaFBr): Motivation**: A high atomic number enhances interaction with protons, potentially leading to higher measurement accuracy in dosimetry. **Aluminum Oxide (Al2O3): Motivation**: Intermediate atomic number, widely used in dosimetry, providing a balance between sensitivity and physical robustness. **Water: Representation**: Utilized as a reference material for its tissue-equivalence properties. **Motivation**: Standard for dosimetry studies, essential for benchmarking and comparison.

Theoretical Models

1. **Bethe-Bloch Theory**: Applied for high-energy protons and Includes density and shell corrections. **Motivation**: Provides robust theoretical foundation for high-energy proton interactions.
2. **Semi-Empirical Model**: Used for low-energy protons. **Motivation**: Enhances accuracy in low-energy regime where the Bethe-Bloch theory alone may not suffice.

Data Sources

- **SRIM (Stopping and Range of Ions in Matter)**: Provided stopping power and range data calculations for various materials. **Motivation**: This study has been widely validated and used in ion interaction studies.
- **PSTAR (Proton Stopping Power and Range Tables)**: validated stopping power data calculations for protons in water and other reference materials. **Motivation**: Standard source for proton stopping power data, ensuring reliability and comparability.

3. Results

Accurate determination of shell correction values is crucial for precise dose calculations in proton therapy (see figure 1), which provides empirical data for a range of clinically relevant light elements (Beryllium, Oxygen, Fluorine, Lithium, Neon, and Aluminum) as a functions of proton energy were provided which contributes to refinement of existing treatment planning systems. The shell correction values exhibit a pronounced peak at lower proton energies, followed by a gradual decline as the proton energy increases. The magnitude of the peak varied across elements, with fluorine exhibiting the most substantial shell correction effect. For all elements studied, the shell correction became negligible at proton energies above approximately 35-MeV. By incorporating these results, clinicians can expect improved dose estimation accuracy, potentially leading to enhanced treatment efficacy and reduced adverse effects.

Figure 2 presents the stopping power data for the key materials used in clinical radiotherapy: oxygen, BaFBr, aluminum, and Al2O3. Calculation data (PSTAR and SRIM-2008) were collected for a wide range of proton energies and compared against the established semi-empirical models used in this study. **Low-energy behavior:** A significant deviation from the PSTAR and SRIM models was observed at low proton energies (<1 MeV) for all the materials investigated. This discrepancy underscores the need for more accurate models in this energy range. **Oxygen and BaFBr:** The stopping powers of oxygen and BaFBr exhibit a pronounced peak at low energies, followed by a gradual decline. **Aluminum and Al2O3:** The stopping power of aluminum and Al2O3 shows a similar trend, with a less pronounced peak compared to that of oxygen and BaFBr. **Data agreement:** At higher energies (>10 MeV), the theoretical calculations agreed well with the PSTAR and SRIM predictions, indicating the reliability of these models in this region. The discrepancies observed at low energies highlight the potential for improved treatment planning by incorporating the presented data into existing models (see Figure 2). By refining the stopping power models, particularly in the low-energy region, clinicians can expect enhanced dose estimation accuracy, leading to optimized treatment plans and potentially reduced side effects, and providing valuable data for model refinement and validation.

Figure 3 presents the stopping power ratios of the key materials used in clinical radiotherapy: water, air, aluminum oxide (Al2O3), and barium fluoride bromide (BaFBr). The ratios were calculated for a range of proton energies from 0 to 250 MeV. The **Water-to-air stopping power ratio:** Exhibits a consistent value of approximately 1.1 throughout the energy range. **Al2O3-to-water stopping power ratio:** Shows a gradual decrease from approximately 1.2 at low energies to 1.0 at higher energies. **BaFBr-to-water stopping power ratio:** Displays a more pronounced peak at low energies, followed by a decline such as Al2O3-to-water. **Water-to-Al2O3 stopping power ratio:** Inversely related to Al2O3-to-water, showing an increase from approximately 0.8 to 1.0. The presented data provide a comprehensive overview of the stopping power ratio behavior for clinically relevant materials. By incorporating these results into treatment planning systems, clinicians can expect improved dose estimation accuracy, leading to optimized treatment plans and potentially reduced adverse effects.

Discussion

In the present study, we investigated the accuracy of dosimeters for measuring linear energy transfer (LET) in proton therapy by evaluating high-Z_{eff} materials (BaFBr) and low-Z_{eff} materials (Al2O3, water). By applying semiempirical models to stopping power ratios for proton energies ranging from 0.01 to 10,000 MeV, we observed that the water-to-air ratio was approximately 1.05 at 10 MeV, the water-to-Al2O3 ratio was approximately 1.12 at 1 MeV, and the BaFBr-to-water ratio was approximately 0.95. These findings align with previous studies and confirm the accuracy of the LET measurements, thus validating the theoretical models used [1].

Our study extends the works of [1-3, 16-22] by validating stopping power calculations with both SRIM-2008 and PSTAR data. We found strong agreement with the results reported in [1-2, 18], which demonstrated the utility of high-Z_{eff}materials for improved LET sensitivity. This is consistent with [14], who validated water as a standard for LET measurements. However, our results also highlight discrepancies at lower proton energies, indicating potential limitations in current dosimeter techniques. This gap was less emphasized in a previous study [13]. This supports the theoretical framework of Bethe-Bloch theory and the semiempirical models used [1-3].

However, the inconsistencies at lower energies suggest that current dosimetric methods may not fully capture the variations in LET, which is a limitation not fully addressed by earlier research [13]. A key strength of our study is the comprehensive comparison of theoretical predictions with empirical (calculations) data from SRIM-2008 and PSTAR, which enhances the robustness of our findings [15-16, 18] and the normalization of data values at 150 MeV, as suggested by [1-2], further validating our results. These limitations include the potential variability in material properties and the reliance on theoretical models, which may not account for all the experimental nuances [15-18].

The findings of this study have significant implications for proton therapy dosimetry, suggesting that high-Z_{eff} materials like BaFBr can enhance LET measurement accuracy. This advancement is crucial for improving radiation therapy outcomes because precise LET measurements are essential for effective dose delivery. This finding supports the use of high-Z_{eff} materials for LET measurements, and alternative interpretations can consider the impact of measurement errors or variations in material properties. Further exploration of these factors could provide a more nuanced understanding of dosimeter accuracy and its implications for clinical applications.

Future research should focus on expanding the range of proton energies and exploring additional tissue types to further validate these concepts using experimental confirmation and measurements to reduce uncertainty and refine LET measurement techniques. Integrating more empirical data with theoretical models can also enhance the accuracy of dosimetric assessments and address the discrepancies observed in lower energy ranges for accurate electronic stopping power data [18-22]. Our results offer valuable insights into the accuracy of various dosimetric materials for LET measurements, which are in excellent agreement with available calculations and theoretical data reported in the literature. The observed differences between high-Z_{eff} and low-Z_{eff} materials emphasize the critical role that material selection plays in dosimetry [1-2, 13]. These insights are expected to guide future research and influence clinical practice by enhancing both radiation therapy and protective measures [13]. Studies have explored the correlation between necrosis, observed through radiographic imaging, and regions of elevated linear energy transfer

(LET) in the brain during proton therapy for CNS and H&N cancer patients and the methods for intensity-modulated proton therapy (IMPT) have been developed to reduce high LET in critical structures near the target volume while maintaining optimal dose distribution [23-24]. The feasibility and potential clinical benefits of LET-guided IMPT optimization have been investigated, indicating possible improvements in treatment outcomes [25].

In conclusion, this study confirms that both high-Z_{eff} materials, such as BaFBr, and low-Z_{eff} materials, including Al2O3 and water, are effective for accurate linear energy transfer (LET) measurements in proton therapy. The consistency between our findings and theoretical models, along with empirical (calculations) data (SRIM-2008 and PSTAR), demonstrates the effectiveness of these materials in improving dosimetry precision, which is crucial for optimizing radiation therapy outcomes.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

There are no competing interests declared by the authors.

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Data Sharing Statement

The data that supported the findings of this study are available at the PSTAR website: https://physics.nist.gov/PhysRefData/Star/Text/PSTAR.html and SRIM database: http://www.srim.org/

Data availability

Third party data:

1. **PSTAR database**: Available from the National Institute of Standards and Technology (NIST) at https://physics.nist.gov/PhysRefData/Star/Text/PSTAR.html.
2. **SRIM database**: Available at http://www.srim.org/.

Extended data:

There are no additional datasets, figures, or supplementary materials provided beyond what is already presented.

References

[1] Setianegara J, Mazur TR, Yang D, Li HH. Dual-storage phosphor proton therapy dosimetry: Simultaneous quantification of dose and linear energy transfer. Med Phys. 2021 Apr;48(4):1941-1955. doi: 10.1002/mp.14748. Epub 2021 Feb 19. PMID: 33525050; PMCID: PMC8058281.

[2] Setianegara J, Mazur TR, Maraghechi B, Darafsheh A, Yang D, Zhao T, Li HH. Quantitative proton radiation therapy dosimetry using the storage phosphor europium-doped potassium chloride. Med Phys. 2020 Oct;47(10):5287-5300. doi: 10.1002/mp.14423. Epub 2020 Aug 16. PMID: 32750155.

[3] Zheng Y, Han Z, Driewer JP, Low DA, Li HH. Theoretical and empirical investigations of KCl : Eu2+ for nearly water-equivalent radiotherapy dosimetry. Med Phys. 2010 Jan;37(1):146-53. doi: 10.1118/1.3271338. PMID: 20175476; PMCID: PMC2801734.
[4] Li HH, Driewer JP, Han Z, Low DA, Yang D, Xiao Z. Two-dimensional high spatial-resolution dosimeter using europium doped potassium chloride: a feasibility study. Phys Med Biol. 2014 Apr 21;59(8):1899-909. doi: 10.1088/0031-9155/59/8/1899. Epub 2014 Mar 20. PMID: 24651448; PMCID: PMC4019811.
[5] Driewer JP, Chen H, Osvet A, Low DA, Li HH. Radiation hardness of the storage phosphor europium doped potassium chloride for radiation therapy dosimetry. Med Phys. 2011 Aug;38(8):4681-8. doi: 10.1118/1.3611043. PMID: 21928642; PMCID: PMC3161504.
[6] Han Z, Driewer JP, Zheng Y, Low DA, Li HH. Quantitative megavoltage radiation therapy dosimetry using the storage phosphor KCl: Eu2+. Med Phys. 2009 Aug;36(8):3748-57. doi: 10.1118/1.3171687. PMID: 19746808; PMCID: PMC2724175.
[7] Alyani Nezhad Z, Geraily G. A review study on application of gel dosimeters in low energy radiation dosimetry. Appl Radiat Isot. 2022 Jan;179:110015. doi: 10.1016/j.apradiso.2021.110015. Epub 2021 Nov 3. PMID: 34753087.
[8] Christensen JB, Togno M, Bossin L, Pakari OV, Safai S, Yukihara EG. Improved simultaneous LET and dose measurements in proton therapy. Sci Rep. 2022 May 18;12(1):8262. doi: 10.1038/s41598-022-10575-4. PMID: 35585205; PMCID: PMC9117334.
[9] Granville DA, Sahoo N, Sawakuchi GO. Calibration of the Al2O3:C optically stimulated luminescence (OSL) signal for linear energy transfer (LET) measurements in therapeutic proton beams. Phys Med Biol. 2014 Aug 7;59(15):4295-310. doi: 10.1088/0031-9155/59/15/4295. Epub 2014 Jul 16. PMID: 25029434.
[10] Inaniwa T, Weichert E, Masuda T, Tanaka S, Matsufuji N, Kanematsu N. Stopping-power ratio of body tissues with updated effective energies and elemental I values for treatment planning of proton therapy and ion beam therapy with helium, carbon, oxygen, and neon ions. Radiol Phys Technol. 2023 Jun;16(2):319-324. doi: 10.1007/s12194-023-00721-6. Epub 2023 Apr 23. PMID: 37087722.
[11] Xu, X., Xie, Y., Zhao, M. *et al.* Investigation on Accuracy of Stopping Power Ratio Prediction Based on Spectral CT. *J. Med. Biol. Eng.* **42**, 845–852 (2022). https://doi.org/10.1007/s40846-022-00761-0
[12] Liu R, Zhao X, Medrano M. Experimental validation of proton physics models of Geant4 for calculating stopping power ratio. J Radiol Prot. 2022 Jun 28;42(2):10.1088/1361-6498/ac7918. doi: 10.1088/1361-6498/ac7918. PMID: 35705062; PMCID: PMC9462414.
[13] Massillon-Jl G. Future directions on low-energy radiation dosimetry. Sci Rep. 2021 May 19;11(1):10569. doi: 10.1038/s41598-021-90152-3. PMID: 34012097; PMCID: PMC8134474.
[14] Zullo, J. R., Kudchadker, R. J., Zhu, X. R., Sahoo, N., & Gillin, M. T. (2010). LiF TLD-100 as a dosimeter in high energy proton beam therapy—can it yield accurate results?. Medical Dosimetry, 35(1), 63-66.
[15] Yanwen Zhang, In-Tae Bae, Kai Sun, Chongmin Wang, Manabu Ishimaru, Zihua Zhu, Weilin Jiang, William J. Weber; Damage profile and ion distribution of slow heavy ions in compounds. *J. Appl. Phys.* 15 May 2009; 105 (10): 104901. https://doi.org/10.1063/1.3118582
[16] Amable, A. K. S., Godsway, B. K., Nyaaba, R. A., & Manson, E. N. (2017). A theoretical study of stopping power and range for low energy (< 3.0 meV) protons in aluminium, germanium, lead, gold and copper solid materials. *Open Science Journal*, *2*(2).

[17] Mahmoud, M. M., Shakir, I. A., & Sleeman, S. Y. (2018, December). Shell Correction in Stopping Power for Protons with Different Ionization Degrees in Al2O3 and ZrO2. In *IOP Conference Series: Materials Science and Engineering* (Vol. 454, No. 1, p. 012019). IOP Publishing.
[18] Akbari, F., Taghizadeh, S., Shvydka, D., Sperling, N. N., & Parsai, E. I. (2023). Predicting electronic stopping powers using stacking ensemble machine learning method. *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions With Materials and Atoms*, *538*, 8-16. https://doi.org/10.1016/j.nimb.2023.02.023
[19] Paul, H., & Sánchez-Parcerisa, D. (2013). A critical overview of recent stopping power programs for positive ions in solid elements. *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions With Materials and Atoms*, *312*, 110-117. https://doi.org/10.1016/j.nimb.2013.07.012
[20] Trujillo-López, L., & Cabrera-Trujillo, R. (2019). Electronic stopping cross section for protons incident on biological and biomedical materials within a FSGO quantum chemistry description. Radiation Physics and Chemistry, 156, 150-158. https://doi.org/10.1016/j.radphyschem.2018.10.013
[21] Usta, M., & Tufan, M. Ç. (2017). Stopping power and range calculations in human tissues by using the Hartree-Fock-Roothaan wave functions. Radiation Physics and Chemistry, 140, 43-50. https://doi.org/10.1016/j.radphyschem.2017.03.005
[22] Akkerman, A., Breskin, A., Chechik, R., & Lifshitz, Y. (2001). Calculation of proton stopping power in the region of its maximum value for several organic materials and water. *Radiation Physics and Chemistry*, *61*(3-6), 333-335. https://doi.org/10.1016/S0969-806X(01)00258-4
[23] Niemierko A, Schuemann J, Niyazi M, Giantsoudi D, Maquilan G, Shih HA, Paganetti H. Brain Necrosis in Adult Patients After Proton Therapy: Is There Evidence for Dependency on Linear Energy Transfer? Int J Radiat Oncol Biol Phys. 2021 Jan 1;109(1):109-119. doi: 10.1016/j.ijrobp.2020.08.058. Epub 2020 Sep 7. PMID: 32911019; PMCID: PMC7736370.
[24] Unkelbach J, Botas P, Giantsoudi D, Gorissen BL, Paganetti H. Reoptimization of Intensity Modulated Proton Therapy Plans Based on Linear Energy Transfer. Int J Radiat Oncol Biol Phys. 2016 Dec 1;96(5):1097-1106. doi: 10.1016/j.ijrobp.2016.08.038. Epub 2016 Sep 1. PMID: 27869082; PMCID: PMC5133459.
[25] Giantsoudi D, Grassberger C, Craft D, Niemierko A, Trofimov A, Paganetti H. Linear energy transfer-guided optimization in intensity modulated proton therapy: feasibility study and clinical potential. Int J Radiat Oncol Biol Phys. 2013 Sep 1;87(1):216-22. doi: 10.1016/j.ijrobp.2013.05.013. Epub 2013 Jun 19. PMID: 23790771; PMCID: PMC4520412.
[26] Berger, J. M., Coursey, J. S., Zucker, M. A., & Chang, J. (2010). PSTAR: Stopping Power and Range Tables for Protons. National Institute of Standards and Technology (NIST). Retrieved from https://physics.nist.gov/PhysRefData/Star/Text/PSTAR.html.
[27] Ziegler, J. F., Ziegler, M. D., & Biersack, J. P. (2010). SRIM – The stopping and range of ions in matter (2010). *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms, 268*(11-12), 1818-1823. Retrieved from http://www.srim.org/.

Figure 1: Graphs showing how the shell correction factor changes with proton energy plots for various elements such as Beryllium, Oxygen, Fluorine, Lithium, Neon, and Al. As shown in Figure 1(a), the shell correction for Beryllium peaks around 5 MeV and gradually decreases with increasing proton energy. Figure 1(b) illustrates the shell correction for Oxygen, with a peak observed at approximately 5 MeV. The shell correction factor for Fluorine, as seen in Figure 1(c), reaches its maximum value near 5 MeV before declining. Figure 1(d) shows the shell correction for Lithium, peaking just below 5 MeV and decreasing with higher proton energies. In Figure 1(e), the shell correction for Neon displays a similar trend, with a maximum at around 5 MeV. The shell correction for Aluminum, as depicted in Figure 1(f), also peaks near 5 MeV and diminishes with increasing proton energy. These insights are essential for accurately modeling energy deposition in different materials at various proton energies, thereby aiding in the refinement of proton therapy dose calculations.

Figure 2: Stopping Power Analysis for Different Materials such as Oxygen, BaFBr, Aluminum, and Al2O3, across a wide range of proton energies. The comparison of theoretical stopping powers (blue and black curves) with empirical (calculations) data from PSTAR (red curves) and SRIM-2008 (green markers) illustrates the accuracy of the models used. Figure 2(a) demonstrates the alignment between theoretical calculations and experimental data for Oxygen's stopping power. As seen in Figure 2(b), the BaFBr stopping power displays deviations at lower energies, indicating potential areas for model refinement. Figures 2(c) and 2(d) show a strong correlation between theoretical and empirical data for Aluminum and Al2O3, respectively. This analysis is crucial for selecting appropriate materials for dosimetry and understanding proton beam interactions.

Figure 3: The graphs display the stopping power ratios for different materials such as water to air, air to water, Al2O3 to water, BaFBr to water, and water to Al2O3, over the proton energy range from 0-250 MeV. Figure 3(a) shows that the water-to-air stopping power ratio remains relatively stable across the proton energy spectrum. Figure 3(b) contrasts the stopping power ratios of Al2O3-to-water and air-to-water, demonstrating their varying attenuation properties. The normalized BaFBr-to-water stopping power ratio, as shown in Figure 3(c), indicates significant material-dependent differences. Figure 3(d) emphasizes the behavior of the water-to-Al2O3 stopping power ratio, which is particularly relevant for dose calculations in medical physics. These ratios provide a comparative analysis of material attenuation properties relative to water, which is essential for normalizing dosimetry readings and ensuring consistent dose delivery in proton therapy.